



SIMATS ENGINEERING

**SAVEETHA INSTITUTE OF MEDICAL AND
TECHNICAL SCIENCES**



Technical presentation

Regular Expressions for Real-World Text Processing

**DSA0302 - NATURAL LANGUAGE
PROCESSING**

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Introduction



What is Regular Expression (Regex)?

A **Regular Expression (Regex)** is a formal pattern language used to identify, search, validate, and extract text from a sequence of characters.

In NLP

Regex is mainly used during **text preprocessing** before machine learning models are applied.

Applications

Email validation

Phone number extraction

URL detection

Token filtering

Data cleaning

Technical Working of Regex

Raw Text



Regex Pattern



Pattern Matching Engine



Matched Text



Extraction / Validation

Steps

1. Input text is given.
2. Regex pattern is compiled.
3. The regex engine scans the text from left to right.
4. Characters are compared with the pattern.
5. Matching text is returned.

Example

Text:

Email me at student123@gmail.com

Pattern:

```
[A-Za-z0-9._%+-]+@[A-Za-z0-9.-]+\.[A-Za-z]{2,}
```

Output:

student123@gmail.com



Regex Components



Character Classes

[A-Z] → Uppercase letters

[a-z] → Lowercase letters

[0-9] → Digits

Quantifiers

* Zero or more

+ One or more

? Optional

{3} Exactly 3 times

{2,5} Between 2 and 5 times

Meta Characters

^ Start of string

\$ End of string

. Any character

\d Digit

\w Word character

\s Whitespace

Example

Phone Number Pattern

\d{10}

Matches

9876543210



NLP Applications



Information Extraction:

Input:

John's email is john@gmail.com

Regex extracts:

john@gmail.com

Data Cleaning:

Before:

Hello!!!@@@

After Regex:

Hello

Text Validation

Checks whether

Email is valid

Mobile number is valid

Date format is correct

Password follows required rules



Advantages and Limitations



Advantages:

- Fast pattern matching
- Reduces manual effort
- Supports data extraction
- Improves NLP preprocessing
- Easy to implement

Limitations:

- Cannot understand text meaning
- Difficult for complex patterns
- Language dependent in some cases



Conclusion



Conclusion:

Regular Expressions are a fundamental NLP preprocessing technique. They efficiently perform **pattern matching, validation, extraction, and cleaning**, making them an essential first step before tokenization, stemming, and machine learning.